

Figure 6.7 Modeling: moving from the real world to idealized world.

problem has been to propose theoretical functions (or laws) that link a value to a probability density. They are known as Normal law, Student's law, binomial law, Fisher's law, etc. It is this additional level of abstraction that makes the notion of probability difficult to grasp because it is far from intuitive.

Figure 6.7 attempts to outline the approach extensively used by analysts to design an analytical procedure. As an example, consider the determination of fat in a food, expressed in g/kg of dry weight. This example is deliberately simple and serves only to illustrate the diagram. The first model consists in saying that lipids correspond to all substances extracted by an organic solvent, such as petroleum ether or hexane. This statement is often false, the model is more complicated and, depending on the food, pretreatments may be necessary before lipid extraction. We finally obtain a value of the lipid measurement denoted X_1 in the figure.

The second model is used to describe the measurement of the amount of dry matter present in the test sample, denoted X_2 . Here again, various water elimination procedures are applicable.

However, behind these two models, which can be considered as chemical, are two statistical models which consist in declaring that each value X_1 or X_2 is made up of two quantities: a true concentration and a measurement error. These new models assume that in the sample to be analyzed, the lipids and dry matter are at constant and uniform concentrations, denoted respectively V_1 and V_2 . There are also the errors made on each measurement, denoted here e_1 and e_2 which form the random part of the model and are more or less *probable*.

In the simple case illustrated here, it would be classic to say that these errors follow Normal probability distribution laws, with zero means. In the Section 8.3 which deals with sampling uncertainty, it is demonstrated that this idea of constant contents is also an ideal that never exists in the real world. Finally, the description of the real world will be reduced to ratioing X_1 and X_2 values.

As already pointed out, experimenters are facing a fundamental problem. When a measurement or an experiment is repeated, the obtained values or results are neither predictable nor identical. Statisticians therefore use the term “random variables” to describe this behavior of the measurement values.

The basic idea behind these few remarks is to remember that a value obtained at the end of an experiment, or an analytical procedure, is only a possible value, “more or less” probable. It is therefore important to keep in mind that the observed measures are only possible values and that repeating measurements would lead to other

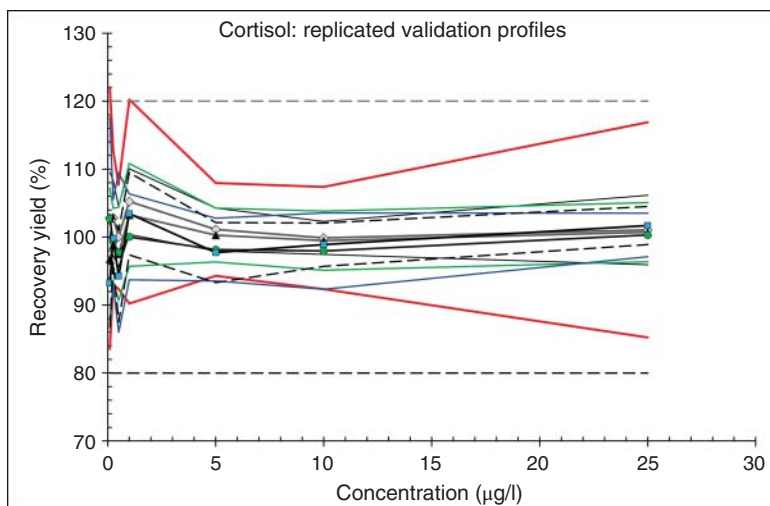


Figure 6.8 CORTISOL – accuracy profiles from four validation studies.

values that are just as probable. The frequential definition of probability implies the idea of an infinite number of trials, so it is impossible to know exactly the probability associated with a value, that would then no longer be random.

On the other hand, the probability density functions developed by statisticians allow us to calculate, from a small number of observed values and for a given probability, a set of possible values contained, for example, in a prediction interval. To illustrate these remarks, Figure 6.8 puts together four repeated accuracy profiles obtained for the same method of analysis of CORTISOL by LC-MS-MS, over a substantial period, and without any modification of the analytical procedure [14].

The method for obtaining and interpreting accuracy profiles has been described in Chapter 5. In this example each profile consisted of three series of three replicates at seven concentration levels; thereafter, identical experimental design was repeated four times for cortisol. The dashed horizontal lines delineate the acceptance interval $\pm 20\%$ and all profiles as thin dashed lines.

The conclusion is that the method is valid over the validation range, whatever the chosen profile. On the same graphic, the β - γ -CTI of the obtained first profile is also reported as two thick solid red lines. It is interesting to remark that almost all β -ETIs are included within the β - γ -CTI obtained the first time. It brings some empirical proof that β -ETIs correctly predicts future measurements.

The construction of the relative uncertainty function from an accuracy profile will be described in Section 7.5.2. Figure 6.9 illustrates the relative uncertainty functions of the method. These functions are clearly affected by the random variability of the profile. For example, the relative uncertainty may vary from 5% to 12% at concentration of 1 mg/ml. Below this concentration, the differences are even greater, but at higher concentrations, the various functions converge. This observation of this

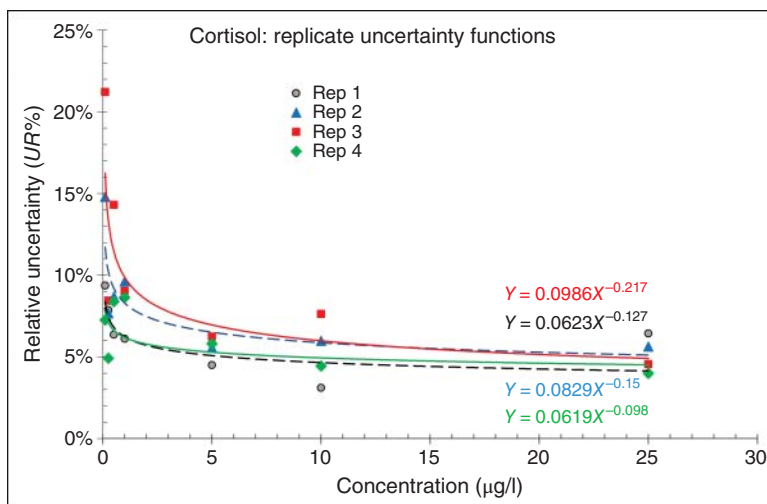


Figure 6.9 CORTISOL – uncertainty functions of four replicated accuracy profiles, denoted Rep 1 to Rep 4.

variability of experimental data must be kept in mind as it is an illustration of the pending problems for the estimation of MU in analytical sciences.

Finally, since the beginning of the nineteenth century, the theoretical tools that are the distribution laws of probability, allow us to predict the probable behavior of the replicates of a measurement method. That is this approach, often called deterministic, that is presented here.

Since the 1970s, several new techniques, based on the intensive computing capabilities provided by computers, also allow the calculation of the probable dispersion of experimental measurements without using a theoretical law. They are sometimes called empirical or without-a-model. Unfortunately, they require different computational methods, such as the Bootstrap, which are not implemented in worksheets and require a specific program for each application [15]. Moreover, the regulatory authorities do not yet consider these innovative approaches as valid, for this reason they are not presented here, without presuming their importance and contribution to the field of analytical sciences in the coming years.

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7

Measurement Uncertainty in Analytical Sciences

7.1 Published Procedures: An Evaluation

The notion of measurement uncertainty (MU) has only recently emerged in analytical sciences. The 2005 version of ISO 17025 noted that “testing laboratories shall have and use procedures for estimating uncertainty of measurement. In certain cases, the nature of the test method may preclude rigorous, metrologically, and statistically valid, calculation of uncertainty of measurement” [1]. MU estimation is now mandatory for accredited laboratories and increasing in numerous application fields, such as environmental, toxicological, and clinical laboratories.

It quickly became apparent that the procedure classically used by other measurement professionals, in particular the Type B approach, was particularly difficult to apply to chemical or biochemical measurements. Since then, many proposals offering complementary, a priori better-adapted solutions have been published. Unfortunately, these proposals are very theoretical; likewise, validation procedures are often sectorial, linked to a specific field of application. It should be remembered, however, that Type A and Type B evaluations are not mutually exclusive and can even be complementary.

Figure 7.1 provides a graphical representation of several published procedures, and Table 7.1 associates the two-letter codes used at the bottom of the diagram with bibliographic references. These documents were selected because they present estimation procedures, official or consensus, specifically dedicated to the analytical sciences. The code AA corresponds to the guide to the expression of uncertainty in measurement (GUM) general procedure, already described in Section 6.2. The most striking example of the various available procedures is provided by the network of the European Directorate for the Quality of Medicines–Official Medicines Control Laboratory (EDQM-OMCL), where five different procedures are proposed for the so-called bottom-up approach only.

This list is not exhaustive, as new proposals are regularly published and/or updated. This results in two main observations:

- It is recognized that different procedures may give different estimates of MU. For instance, with the LEAD dataset, relative MU values ranging from 1 to 6 were obtained depending on the sources of uncertainty considered and the calculation method applied [7].